

TAHMID AHMED TALUKDER

Computer Science and Engineering Undergraduate | United International University

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Professional Summary

CSE undergraduate and published co-author with hands-on experience building AI-enabled web applications, real-time collaboration systems, desktop software, and research prototypes. Technical work spans Laravel and FastAPI, Java and JavaFX, PyTorch and PyTorch Geometric, NLP, multimodal learning, database-backed systems, and human-centered design. Earned First Runner-Up and Top 10 recognition at the UIU CSE Project Show.

Education

United International University , Dhaka, Bangladesh	2022–2026
Bachelor of Science in Computer Science and Engineering CGPA: 3.35 / 4.00	
Relevant Coursework: Machine Learning; Data Mining; Human-Computer Interaction; Advanced Object-Oriented Programming; Software Engineering	
Academia , Gulshan, Dhaka, Bangladesh	2020–2022
GCE Advanced Level (A Level)	
Mastermind School , Uttara, Dhaka, Bangladesh	2013–2020
GCE Ordinary Level (O Level)	
International Hope School , Gulshan, Dhaka, Bangladesh	2010–2013
Primary Education	
Thomas Buxton Primary School , Whitechapel, London, UK	2006–2010
Primary Education	

Technical Skills

Programming: Python, Java, C, C++, PHP, JavaScript, SQL, HTML, CSS

Frameworks and Libraries: Laravel 11, FastAPI, PyTorch, PyTorch Geometric, JavaFX 17, FXGL, SBERT, faster-whisper

Databases and Retrieval: MySQL, ChromaDB

Tools and Platforms: Git, Maven, Figma, FFmpeg, Laravel Reverb, WebSockets, Jitsi Meet, OpenAI API, OmegaConf, YAML

Methods: Object-oriented design, REST APIs, relational database design, retrieval-augmented generation, NLP, multimodal learning, graph neural networks, human-centered design, usability testing

Software Engineering Projects

Cognisense	Jan 2026–Present
<i>Laravel 11, FastAPI, MySQL, ChromaDB, WebSockets / Founder and Developer</i>	
- Built a full-stack AI skill-assessment platform with 13 scenario-based categories and voice, video, and text responses; connected browser capture and Laravel storage to a FastAPI, FFmpeg, and faster-whisper transcription pipeline. - Implemented hybrid response scoring with SBERT embeddings, cross-encoder natural language inference, LanguageTool, and fluency features, including topic validation and off-topic score caps. - Integrated a ChromaDB-backed retrieval-augmented chatbot, real-time collaboration through Laravel Reverb, role-based authentication, notifications, and dynamic PDF CV and certificate generation.	

Project Demo

AspireIELTS	Jan 2025–Present
<i>PHP, MySQL, JavaScript, OpenAI API, Jitsi Meet / Founder and Developer</i>	
- Engineered database-driven IELTS mock testing across Listening, Reading, Writing, and Speaking, including timed sequencing, word counts, answer persistence, scoring, feedback, and progress dashboards. - Integrated GPT-powered writing evaluation and learner guidance, live Jitsi speaking assessments, admin monitoring and reporting, automatic certificate email delivery, and dark/light themes.	

Project Demo

ArcadiaFX	Oct 2025–Present
<i>Java 17, JavaFX 17, FXGL, Maven, MySQL, WebSockets / Founder and Developer</i>	
- Built a modular desktop gaming hub containing 20 functional games, with separated controllers, models, services, and utilities; unified game launching; reusable asset and audio management; persistence; and theme controls. - Added SHA-512 authentication, MySQL-backed player data, WebSocket chat, OpenAI-powered game discovery, feedback and PDF performance reports, lazy loading, caching, and background processing. - Earned First Runner-Up in the Advanced Object-Oriented Programming category at the UIU CSE Project Show.	

Project Demo

Research and Human-Centered Projects

CASH-HyperLBS: Resolving Sign Ambiguity in Higher-Order Drug Interactions via Calibrated Hypergraph Learning

Python, PyTorch, PyTorch Geometric, SapBERT, Llama-3 / Data Mining Course Project

- Co-designed a signed-ambiguous hypergraph framework for higher-order adverse drug-event detection using HODDI/FAERS and outcome-independent mechanistic features from CYP450, Reactome, ChEMBL, ATC, and TWOSIDES.
- Specified a modular PyTorch Geometric pipeline for SapBERT drug harmonization, three-channel message passing, Monte Carlo structural-frustration estimation, contrastive pretraining, and confidence-weighted LLM distillation.
- Defined reproducible prospective evaluation using a strict temporal split, AUROC, AUPRC, and precision-recall analysis without claiming unreported performance results.

Neuro-Symbolic Multimodal AI for Interpretable Depression Screening with Proxy Anxiety and Psychological Distress Indicators

Python, PyTorch, Hugging Face Transformers, MentalRoBERTa, Wav2Vec2 / Final Year Design Project, Jan 2026–Jun 2026

- Contributed to a five-member FYDP that fused Wav2Vec2 audio and MentalRoBERTa text through a 15-concept bottleneck for interpretable PHQ-8 severity estimation and explicitly labeled proxy anxiety indicators.
- Implemented an LLM concept-annotation pipeline for 275 E-DAIC transcripts and 15 concepts, producing 4,125 structured annotations; pretrained MentalRoBERTa on 27,977 four-class mental-health posts and co-trained the concept-bottleneck configurations.
- Ran ablation and $\lambda \pm 0.1$ sensitivity analyses; the full system achieved test RMSE 5.53, MAE 4.03, macro-F1 0.64, and PR-AUC 0.69 on 56 E-DAIC test sessions.

PScnv-PoNMatch: Adaptive Panel-of-Normals Selection and Confidence-Aware CNV Detection for Targeted Sequencing

Computational Genomics, Panel of Normals, Ridge Regression, CNV-Sculptor / Bioinformatics Course Paper, Jun 2026

- Co-authored a CNV-detection framework for unmatched targeted sequencing that combined Panel-of-Normals normalization, IQR-guarded diploid-anchor selection, Ridge denoising, cellularity-adaptive Z-segmentation, and purity-aware calling.
- Benchmarked PScnv-PoN against XHMM, CNVkit, CODEX2, and standard PScnv on 75 simulated multi-cancer profiles spanning five tumor-purity levels and 122 sparse target regions.
- Achieved precision 0.7188, recall 0.8169, and F1 0.7314—a 42.9% relative F1 gain over standard PScnv—while retaining F1 0.5711 at 20% tumor cellularity ($p < 0.000001$, Wilcoxon signed-rank test).

Shathi: A Voice-First AI Caregiving Companion

Jun 2026

Figma, User Research, Usability Testing / HCI Course Project

- Collaborated in a five-person team to design a Bangla-first caregiving companion for female informal caregivers, translating six 20-40 minute semi-structured interviews into requirements for voice access, privacy, trust, and low-friction navigation.
- Helped develop and refine low- and high-fidelity Figma prototypes for voice and text input, medication reminders, routine history, privacy settings, and emergency guidance.
- Evaluated the prototype with five participants across four task-based think-aloud activities, then improved microphone visibility, persistent navigation, Bangla safety messaging, and the voice/text toggle.

Multimodal Mental Health Prediction

Jan 2026–Present

PyTorch, Transformers, wav2vec2/HuBERT, Integrated Gradients / Research Project

- Designed an explainable multi-task, multimodal framework using E-DAIC interview text and audio to predict depression (PHQ-8), PTSD (PCL-C), and weakly supervised anxiety with Transformer text encoders, speech representations, and cross-attention or gated fusion.
- Implemented modality dropout, missing-modality inference, uncertainty-weighted objectives, and token/time-window attribution; defined MAE, RMSE, F1, AUROC, ablation, calibration, and faithfulness evaluation protocols.
- Prepared an IEEE-format research manuscript documenting the methodology and experimental design.

Project Demo

Additional Projects

DormPilot

Jan 2026–Present

Full-Stack Web Application, Relational Database, PDF Reporting / Co-Founder and Developer

- Co-built a role-based hostel management platform for administrators, supervisors, residents, and guests, covering admissions, room allocation, task assignment, complaints, room changes, announcements, and status tracking.
- Created dashboards for occupancy, vacancies, complaint trends, and maintenance costs, plus automated PDF reports, resident-supervisor chat, and notifications.

PHP, MySQL, JavaScript, AI Chatbot / Co-Founder and Developer

- Co-developed a global education guidance platform with country-based university search, database-backed requirements and eligibility data, and a schema for universities, countries, and student applications.
- Integrated an AI chatbot for university selection, scholarships, and admission pathways; added offer-letter request workflows, optimized data retrieval, and secure form submissions.

Publication

Co-author, “Quantum Computing Applications in High-Speed Signal Processing for EEE Systems,” *International Journal of Engineering, Business and Management*, vol. 8, no. 4, pp. 48-57, 2024. doi:10.22161/ijebm.8.4.7

Awards and Leadership

- **First Runner-Up**, Advanced Object-Oriented Programming, UIU CSE Project Show - ArcadiaFX; **Top 10 Project**, Software Engineering Lab category, UIU CSE Project Show - Cognisense.
- **IDP Education Roadshow Volunteer**, Dhaka | Feb, Mar, and Jul 2024; Jan 2025. Supported attendee inquiries and coordinated interactions among prospective students, parents, and university representatives across four events.

Certifications and Languages

Grameenphone Academy Professional Skills Development Program | Mar 2026: Smart CV; Art of Communication; Corporate Presentation Skills; LinkedIn 101; Sharpen Your Interview Skills

British Council Book Reading Competition | 2015, 2016, 2017: Certificate of Achievement

Languages: English (C1), Bengali (C1), Arabic (A2)